Liberation engineering for resistance networks

A Manifesto of the Hoapa Project

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Overview

The problem, in the light of a *fundamental theorem* of communication systems

- Context for the fundamental theorem
- The fundamental theorem of "world-describing communication systems"
- The current situation of humanity, as a diagnosis
- The fundamental theorem in human history
- The "world-exploiting economy" viral state

The solution, a type of communication system

- The legitimacy of the recovery procedure
- The population's readiness
- Implementation
- Effects on science
- Effects on the economy

The Problem — a "virus" in communication systems

Our goal here is to organize the liberation of humanity. In general, but by specifically targeting institutions monopolizing a certain relation to the world and Nature. Our strategy is to clear up the perceived complexity with a sharp diagnosis of the situation, revealed by a simple observation about communication, legitimacy and trust. The solution should then become self-evident.

Theory

To see the full picture and be able to mobilize all relevant concepts for our purpose we'll frame the fundamental theorem in a very general context. We'll consider 4 levels:

- the agents : humans
- the communication system : the entire communication infrastructure of humanity

- the total system (of which the communication system is a component) : humanity
- the context (in which the total system operates) : the world

In this text we're particularly interested in viewing the communication system as having the function of understanding and describing the world/context, for the benefit of the total system—this is where the problem lies.

The first principle of engineering states that "everything is a trade-off". You can't maximize all properties of a system at will, Nature asks you to make decisions. For example, an engine is either very fast or very sturdy.

Our core principle is in the form of a such a trade-off, at the level of the communication system, between:

- quality of information and fluidity of its transmission
- global recognition of expertise

Say we are a communication system designer starting from scratch, and we're given the requirement that the system must be able to fix the legitimacy of selected agents. This is actually a very strong specification and it forces us into certain architectures and mechanisms. Those in turn necessarily have costs in terms of information transfer (availability, clarity...) and induce degradation in the information itself (distortion, fragmentation, reduced scope...). Crucially, this "fixing of legitimacy" becomes a sort of game which interferes with the original world-describing function of the system, along with many more subtle effects.

To make a long story short: it creates a narrow pyramid while optimal information creation and transfer requires a wide flat surface—a rugged and fragmented pyramid of privileges rather than a smooth continuous plane of engagement.

The opposite side of the issue is even more dramatic for the total system: very few agents are involved in this special relationship to Nature centered around its exploration and understanding, which constitutes the backbone of any agent's life, regardless of function in the total system. This is a form of mutilation, constantly repeated at the individual level, and faithfully echoed at the global scale.

What is at stake is mainly:

- The breadth of the scope of the world-describing apparatus and the elements on which it focuses
- The quantity and importance of the assumptions implicated in the evaluation of Nature
- The trust with which the results can be received by the population

The fundamental theorem:

> *"In order to be maximally efficient, a world-describing communication system embedded in a world of agents must not engage in the selection and specialization of agents. It must not designate representatives, adopt hierarchical structures or materially reward information creation. All of those must only be initiated locally by the free agents, individually or in certain autonomous and decentralized networks."*

We'll see in the second section of the text, the solution, how the last part can be readily achieved.

Biological metaphor

The kind of systems this theorem applies to are necessarily of high complexity, analogous to biological organisms rather than deterministic machines. Systems failing to follow the theorem's principles

won't see an obvious breakdown of the intended function of the communication system: even if suboptimal, there will still be a form of understanding and description of the world emanating from it. But problems will emerge in different places and in various forms in the total system. The total system which will in turn react to the situation, further complicating the diagnosis.

It's as if today's humanity went to the doctor:

> — Doc, I have constants bouts of war and I'm afraid it will burst into a world conflict again. My trust levels are very low and my democracy hurts. I think I have had genocide again. And I'm addicted to money and technology, it's all I seem to care about... I neglect friends and family.
>
> — You have a university problem.
>
> — ...Really? I thought that was the one thing that was keeping me alive...
>
> — You see, your universities actually do perform the vital function of creating and refining information about your world, which is why you think it's your protector in your misery. But, like a kidney in a human body, if it doesn't do its job well enough, then the whole body suffers from poisoning. This is actually related to a central authority virus, which hoards valuable information coming from the universities on one side, for its own sinister purposes, and spews out toxic messages into the information-stream to prevent the population from reacting to it.
>
> — But what about Einstein's relativity? My universities are working, I can feel it! And they're much better for me than the organized religions I had before.
>
> — You're actually quite late in your scientific development, compared to healthy societies. Just look at your scientific breakthroughs: very far apart in time, coming from a very small sample of your population, who usually have to miraculously survive the selection process and hierarchies to get their information recognized. And more than 100 years after relativity, the information is only marginally widespread. Your population needs to *know*, not *know of*. You still produce healthy information, but your organism as a whole is not healthy. And if universities are an upgrade it mostly benefits your central authority virus for the moment, while you seem to have lost some dignity. You need to flatten your science process and involve your population again.

To gain some perspective, let's look at the patient's history.

History

In the light of the above theorem and the flat/pyramidal metaphor, we can distinguish 4 phases in the history of human world-describing communication systems:

- An unknown original state
- A flat surface (mostly Paleolithic) — the age of mythology
- Pyramids via organized religion — the age of temples
- Pyramids via science — the age of universities

To comment on these:

- There is a major misunderstanding regarding the function of ancient mythology: it is not an authoritative worldview but a vehicle for all relevant information the individual may need to create their own, in symbolic form—but most of the mythology we have access to have been gravely distorted in the next phase.
- It is also inappropriate to say organized religion only mobilizes superstition. There is a lot more to the "hacking" of mythology and spirituality, particularly regarding the social cohesion and

solidarity aspects—present in the words of the prophets, but inexorably corrupted by religious institutions.

- The scientific method is a major breakthrough in the history of humanity, but unless it operates on a flat surface, it is incomplete and in many ways dangerous—this is where the solution to most of our current problems lie.

The currently dominant narrative regarding world-describing communication systems is inherited from the enlightenment, the architects of the transition between the age of temples and that of universities. It explicitly affirms the main antidote to superstition-based tyranny is knowledge. And implicitly that the only way to knowledge is through *selected and specialized representatives organized in hierarchical structures: the modern university.*

With technology replacing make-believe as the main output, the universities have become exactly what they were conceived to resist, only in a different form: a tool for central authorities to establish a firmer grip on populations. With the added perk that now they can exploit Nature to a dramatic new degree.

The dual concept: the central authority virus

The naïve view of long-term human history assumes there was no economy before civilization and that mythology has been replaced by facts. Both are wrong, the economy started with tool making and there is still a mythology today, usually called culture—now mostly advertising.

If we posit that communication systems mediate the relationship between agents and their world in the conceptual realm, then we can find its counterpart in the physical realm: the economy. While the communication systems create, refine and transport ideas, the economy does the same with things extracted from the world.

We can repeat the theorem: a total system whose economy is not flat but pyramidal exposes itself to degeneracies. Except, the physical consequences of a diseased economy are far more dramatic, for the individual agents, the total system and the world. All will be severely abused. Whereas a peaceful and sustainable path to societal plenitude exists, precisely what our liberation networks can delineate.

We have a tangible trace of this duality in human history: the temple and the palace. Sumer is considered to be the first civilization. While it started with religion-oriented temples as central authorities, they were quickly joined by war- and economy-oriented palaces. Their well-balanced dance of exploitation lasted for centuries. And following regular patterns of colonial growth, it spread across the globe.

At the opposite end of the spectrum we have the example of African Egalitarianism: a system which respects both theorems. It was applied by the most ancient cultures of humanity, taking their roots in the Paleolithic, and barely surviving in small isolated pockets today. Their rich abstract life was a form of decentralized global conversation. Their stable and balanced economy was a vast network of consent, based on open and fluid group membership, superposition of land ownership and solidarity-inducing gifts.

If we follow the newest orientations of archaeology, we're led to believe those systems were probably the norm for humanity in the Paleolithic, but they started to show signs of infection during the Neolithic, and fell prey to the virus with the first civilizations. However, the healthy processes were still active at this time, and one of those early civilization was actually an extension of them, proving that they also can flourish into large structures: the Indus Valley. Based on exceptional conditions for food production, its economy was well distributed via tokens of consent (seals), and its infrastructure was optimized for the commons (notably formidable city-wide water systems).

While its contemporaries all spread through successive conquests and collapses, it quietly faded from existence by returning to a more robust agricultural lifestyle when changes in the weather ended the production of massive surpluses.

Later, this region will see the arrival of the central authority virus in the person of Darius the Great, starting a long series of power struggles which have now become the tension between the modern states of Pakistan and India—often described as the most dangerous "nuclear flashpoint" in the world.

The Solution — a healthy communication system

Next I will propose a procedure for humanity to recover from the dual imbalance considered above. Following the biological metaphor, it's as if the body had forgotten how to produce the "healthy interactions" and "adequate tasks" which should be the bases of its communication system and economy respectively. One needs to properly formalize what they are, how they work, and why they're efficient.

Then, this information needs to enter the communication system and receive legitimacy, the currency by which the signal will be evaluated. Without legitimacy, the population won't pick up the new model, and the healthy process won't restart.

In brief, the type of communication system suggested by the fundamental theorem is uniquely placed on its spectrum, making it a distinguished point in its category. It has the same kind of legitimacy as the center of a circle: it is special because it is balanced and given naturally, while any other chosen point on the disk would necessarily be arbitrary—and *this* choice would raise legitimate questions.

I have given my best effort to demonstrate the deep harmony between these ideas and what we have observed from Nature, from philosophy and mathematics to physics and biology. And while I have received praise for my work in the rare cases I was given an opportunity to present them to scholars, none of the people I talked to grasped their potential and significance, or wanted to act on it. I hope this text will find an audience that does.

We'll try to answer those questions:

- Do we need to trick the total system to get it to recognize the legitimacy of the recovery procedure?
- Are the agents ready to accept it?
- How is it implemented?
- What effects can be expected on science?
- What effects can be expected on the economy?

I'll be brief and invite interested readers to contact me or read my other texts for a more refined analysis: doc2cog@proton.me | `https://codeberg.org/doc2cog`

Do we need to trick the total system to get it to recognize the legitimacy of the recovery procedure?

Because the current network of world-describing institutions uses the galilean scientific method for its operations, we should be able to get it to recognize the significance and potential of the type of

communication system suggested by the fundamental theorem above. There are several meaningful questions we can ask Nature, to which the answer will be the healthy liberation networks. The difficulties lie in:

- Some deeply ingrained assumptions regarding the indispensability of a form of global control in any human endeavour. To the contrary, the liberation networks are autonomous, decentralized and rely on the "natural" power of conversation when all participants have been put on an equal footing (the last part is the tricky bit, where all institutions automatically fail for structural reasons).

- The limited scope and technology-oriented focus of the current apparatus: the demonstration of the legitimacy of liberation networks consists in making a series of correspondences between entire domains of knowledge (quantum mechanics and relativity, concurrent programming, generative linguistics, relational biology, perspectivism in philosophy...), via a common mathematical model equipped with amazing information-processing capabilities (sheaf toposes). We can go further and claim the healthy networks actually generate and structure the evolution of all the systems included in the correspondence. Because of their breadth, together they yield a generalized theory of evolution (see my text "An introduction to Grothendieck's general theory of evolution" and the favorable comments by Fernando Zalamea). All this feels alien to a structurally fragmented and disconnected academic world.

This "magical" mathematical model is like an envelope one puts on almost any structure, to enrich and clarify vision. It is extremely abstract and powerful, with roots in pure geometry and logic. It has silently revolutionized mathematics. It also provides alternative ways in which to engage with information collectively, so that it is filtered and refined without having to sacrifice a vast part of the collective's participation.

Here are the core principles of this model, the ingredients which make it so good at extracting meaning from any sort of landscape:

- Coordination: all "local pieces of truth" are *respectful* of one another — a "global point of view" can only be set up if the local data composes harmoniously

- Contextuality: all appreciation of truth depends on *context*, and unless specified it is flexible, fuzzy and patient — non-binary, capable of enduring incoherent steps and recover

- Meaning: all conflicting data are not perceived as obstacles to totalization but on the contrary as sources of *relevant information* — totalization is not an objective, only coordination

(Does that remind you of something?)

Are the agents ready to accept it?

The popular protests of 2019 were a beautiful surge of healthy rage and collective hope, when 40 incarcerated populations took to the streets to demand change. Without any coordination or planning, globally or locally, all those movements showed the same respect for the core principles just mentioned above. They were my inspiration for designing and studying this type of system. I believe it's the adequate structure into which their energy must flow to become a constructive force of liberation.

How is it implemented?

In what follows, the reader must be mindful of how the healthy processes/liberation networks evolve: they take a lot of time and trials and errors before they reach maturity. Again, they are not machines you turn on and off, but more like organisms going through life cycles. They need to be babies, and it's not easy to imagine what they will look like as adults (and it would terribly presumptuous for anyone to claim to know what millions of people could come up with when properly organized).

If they respect the following guiding principle, they can grow large while preserving respectful coordination, contextuality and meaning formation:

> *"The system must only know one way to interact with its participants. Alternatively, participants are indistinguishable in the eye of the system."*

In this way the nice properties coming from abstract mathematics find themselves encapsulated in a concurrent computational system (in jargon: *sheaf ∞-toposes* are implemented in the *actor model*). Both are grounded on embracing locality and communication, and the rejection of a global state. Here the units of computation are the brains of the participants, generating a truly distributed cognitive system. (You will remark how current science is grounded on the opposite assumption: its knowledge is structured around distinguishing between scientists and non-scientists, and then among scientists themselves.)

In practice this principle yields a "conversation of conversations".

An adequate model for a single instance of a "conversation inside a conversation" can be found in recent innovations in participative democracy, the citizen assemblies or citizen conventions: a dozen people or so are randomly selected from the entire population and given ample time to discuss a particular issue, before giving a report of their discussion, written by consensus. (You can see that there is only one path through this system: { being selected randomly Y/N? If yes and willing to, participate and decide what goes in the report }. The selection makes the conversation possible, but the randomness doesn't distinguish between potential participants.)

The "conversation of conversations" level of the system is mostly working on building correspondences between concepts (*homotopies*) and extracting meaning from possible contradictions (*singularities*).

In essence, the true intent of the system is to reinvent a technical language, as a gift to humanity. Because language is the main tool we use to engage with the complexity of the world, it should be as sharp *and* accessible as possible, so as to lower the cost of individual and shared understanding. This is how this system reaches perfection for its purpose of maximizing quality of information and fluidity of transmission.

Through these seemingly insensitive mechanisms, another natural phenomenon at the heart of human cognition is also reactivated: caregiving, known as attachment, or love.

What effects can be expected on science?

The scientific method will be extended to the collective. While the Galilean method perfectly captures what the relation between a scientist and Nature must be for the former to properly ask question to the latter, it doesn't say anything about how scientists should interact. Historically, it has been another polite distraction for the upperclass. Today, the peer-review system and editorial infrastructure are dysfunctional ad hoc extensions. They should be replaced by a network of implementations of healthy communication systems.

The main changes would be:

- The system will know how to filter and utilize input without selection of agents, so all voluntary participants can get involved (again: greater scope, less assumptions, more trust, but also: more creativity, pedagogy and simply power in numbers)
- Education will be freed from having to select among children, and can focus on development, customized for each cognitive profile (a generalization of how Finland reinvented education in the period 1970-2000)
- The population, through decentralized autonomous networks, will focus attention on what matters to them and their well-being—as opposed to investors, directors and generals. The world-describing "organ" of the total system will recover its purpose and facilitate the harmony between the agents, the total system and its context—or between humans, humanity and the environment—as opposed to the mere monitoring of maximal exploitation.

What effects can be expected on the economy?

The end of the current debt-based forced labor, across the globe. The time liberated can then be reinvested in what matters to people. The monumental stress of capitalism can be released and a new relation to Nature can begin.

In practice (educated guesses, for inspiration): local autonomous open-source-style projects with distributed management, open workshops and kitchens, reservation systems replace currency, a global inventory to reduce production and encourage upcycling, remodelling of the landscape for minimal human imprint, the philosophy of permaculture generalizes, especially in the traumatized tech world...